Thermodynamic Quantum Logic with Floquet-Buffered Reservoir Coupling

Thermodynamic Quantum Gates Project

August 8, 2026

Abstract

Autonomous quantum thermal machines provide a route to computation in which logical states are encoded in thermodynamic variables and processed by heat currents rather than externally clocked unitary gates. We reconstruct the weak-coupling thermodynamic-neuron model for NOT, NOR, majority, and XOR logic, using the virtual-temperature mechanism of the baseline theory as the Markovian reference. We then extend the numerical programme toward structured reservoirs by validating a single-qubit non-Markovian benchmark and by introducing a periodically driven Floquet buffer between a logical qubit and its bath. The buffer is assessed operationally through output-state distinguishability and an integrated drive-work proxy. In the present parameter set, the buffered architecture increases the final trace distance from 0.1353 to 0.1818, while a broader screening sweep finds positive distinguishability gain in 14 of 15 tested configurations. These results support a cautious next step: a minimal Floquet-buffered thermodynamic NOT prototype, before any claim of a full multi-qubit non-Markovian thermal logic gate.

1 Introduction

Thermodynamic computing asks whether physical relaxation can be used directly as a computational resource. In the thermodynamic-neuron model, logical inputs are encoded as inverse temperatures, a collector induces heat currents through a finite output reservoir, and the final output inverse temperature is decoded as a logical value. The main appeal of this architecture is that computation is autonomous: the system relaxes toward a logical answer without an externally prescribed sequence of gates.

The present work has two aims. First, we reproduce the analytical Markovian thermodynamic-neuron response of the baseline model, thereby establishing a reference for logic fidelity and dissipation. Second, we formulate the next extension: replacing ideal memoryless reservoir contact by structured-bath dynamics, and testing whether a driven Floquet buffer can improve output distinguishability before the logical subsystem loses information to the environment.

2 Markovian Thermodynamic-Neuron Model

2.1 Thermal Occupation and Virtual Temperature

The elementary thermal nonlinearity is the two-level occupation

$$g(\beta\epsilon) = \frac{1}{1+\exp(\beta\epsilon)}, \tag{1}$$

where ϵ is the qubit gap and β is the inverse temperature. For the three-qubit collector used in the NOT gate, the relevant virtual inverse temperature is

$$\beta_v = \frac{\epsilon_0}{\epsilon_z}\beta_0 - \frac{\epsilon_1}{\epsilon_z}\beta_1, \qquad \epsilon_z = \epsilon_0 - \epsilon_1. \tag{2}$$

Thus the input temperature β_1 controls the sign and magnitude of the effective thermal bias seen by the output reservoir.

2.2 Reset-Model Currents

In the weak-coupling reset description, the collector and modulator currents are

$$j_C = \mu\epsilon_z \left[g_z(\beta_z) - g_z(\beta_v)\right], \tag{3}$$
$$j_M = \mu'\epsilon_z \left[g_z(\beta_z) - g_z(\beta_r)\right], \tag{4}$$

with $g_z(\beta) = g(\beta\epsilon_z)$. The finite output reservoir evolves according to

$$\dot{\beta}_z(t) = -\frac{\beta_z(t)^2}{C}\left[j_C(t) + j_M(t)\right]. \tag{5}$$

The steady condition $j_C + j_M = 0$ gives the bounded output β_z^∞, which is the decoded logical response.

3 Baseline Logic Reconstruction

The reconstructed Markovian response includes the virtual-temperature regimes, NOT transfer curve, error–dissipation trade-off, NOR surface, three-input majority response, and an XOR network assembled from thermodynamic-neuron subgates. The role of this stage is not merely illustrative; it fixes the baseline against which all non-Markovian or Floquet-buffered extensions must be judged.

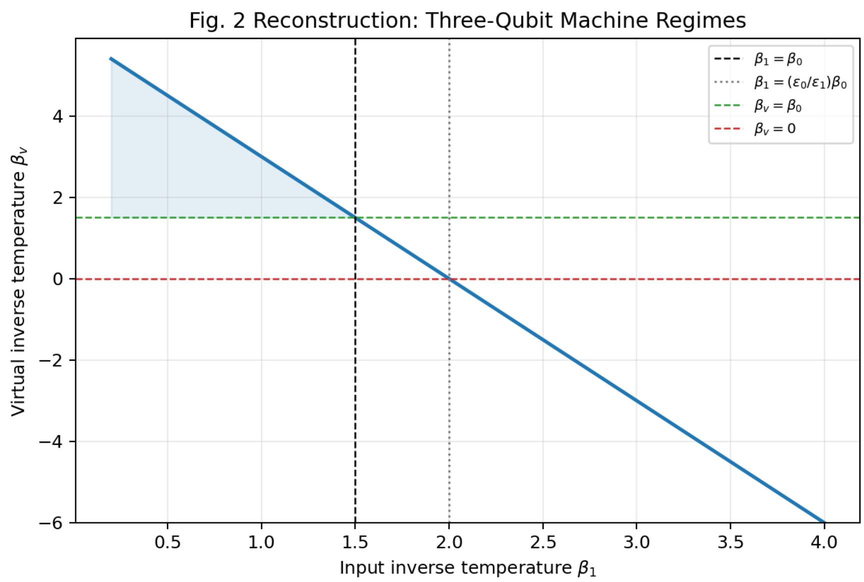

Figure 1: Virtual-temperature regimes of the collector. Sweeping the input inverse temperature changes the virtual thermal bias and therefore the direction in which the collector drives the finite output reservoir.

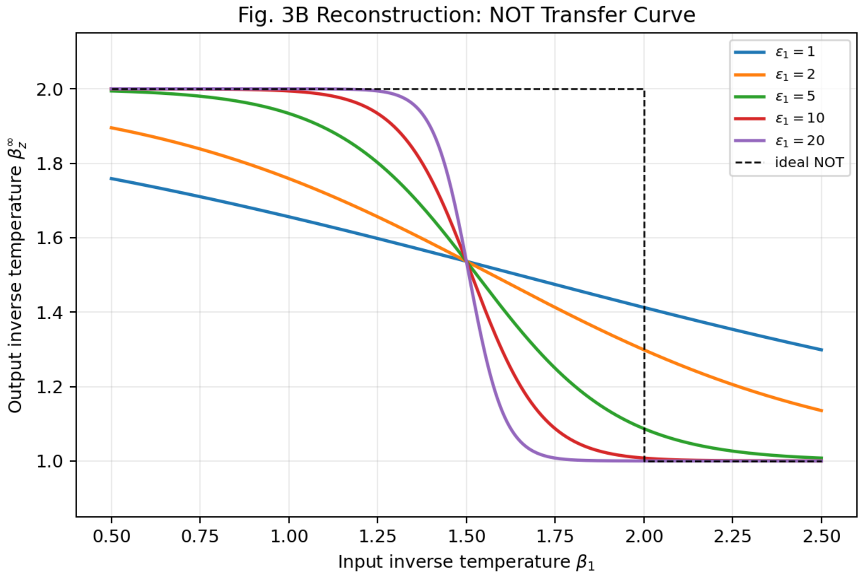

Figure 2: Thermodynamic NOT response. The steady output inverse temperature switches between the two logical temperature levels as the input inverse temperature is varied. Increasing the collector energy scale sharpens the transfer curve.

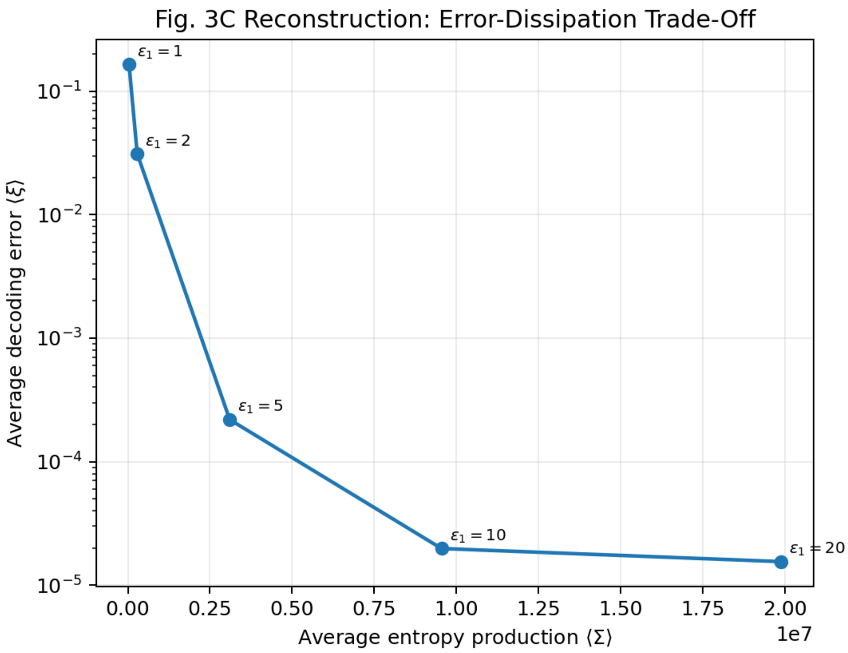

Figure 3: Error–dissipation relation for the reconstructed NOT gate. Larger energy scales suppress decoding error, but the entropy-production cost increases strongly.

Table 1: Representative error–dissipation data for increasing collector energy scale.

ϵ_1	Average error	Average entropy production	Hot-input output
1	1.6477×10^{-1}	3.7266×10^{4}	1.6561
2	3.1114×10^{-2}	2.8675×10^{5}	1.7588
5	2.1882×10^{-4}	3.1029×10^{6}	1.9338
10	1.9770×10^{-5}	9.5496×10^{6}	1.9942
20	1.5480×10^{-5}	1.9882×10^{7}	2.0000

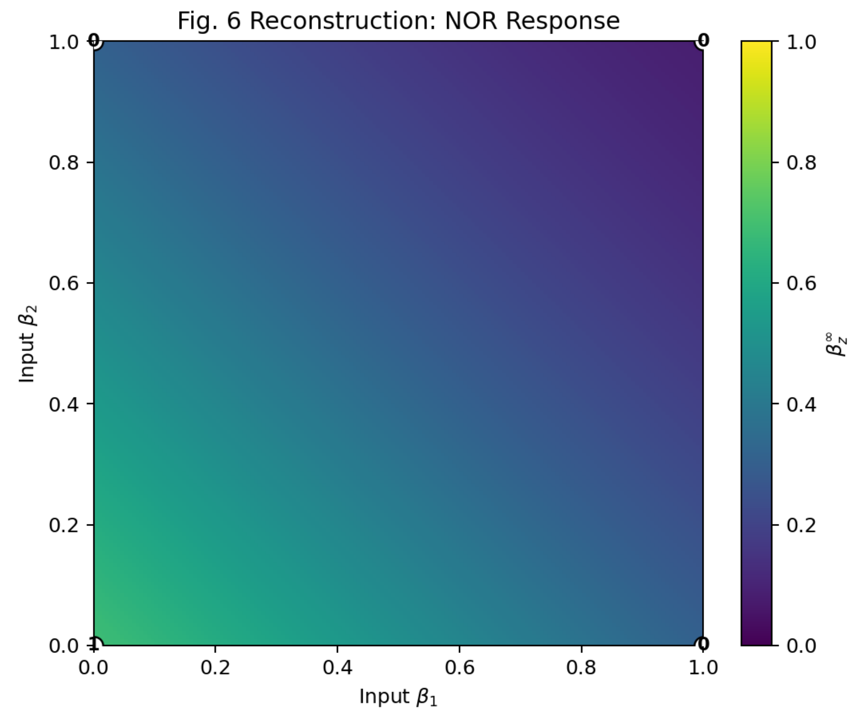

Figure 4: Thermodynamic NOR response. The high-output region occurs only near the low-low input corner, as required by NOR logic.

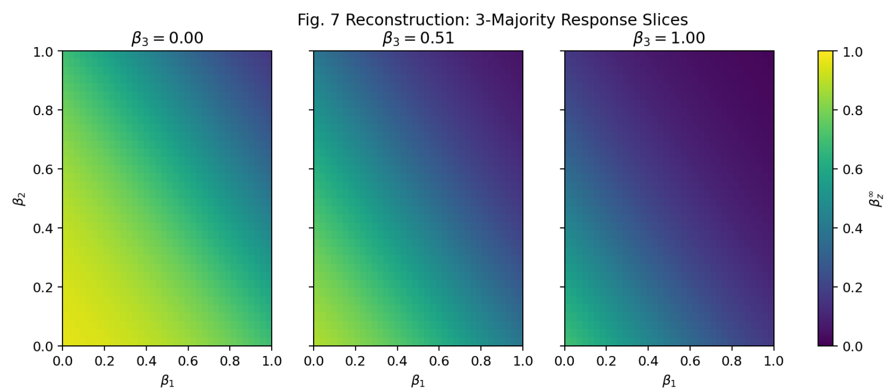

Figure 5: Three-input majority response shown through two-dimensional slices. The response is consistent with a thermodynamic perceptron whose virtual temperature is a linear score of the input temperatures.

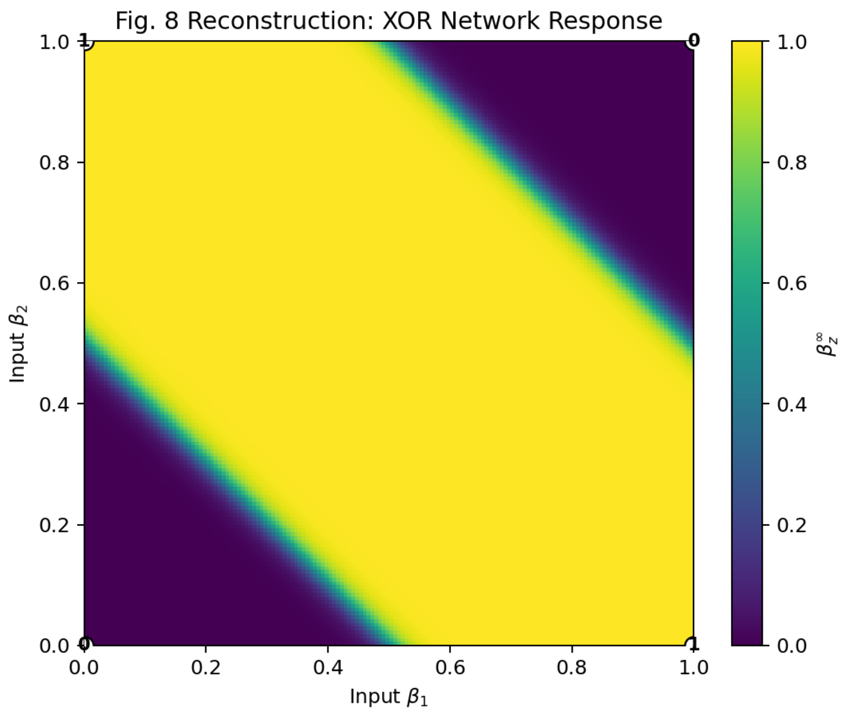

Figure 6: XOR response obtained by composing thermodynamic-neuron subgates. Since XOR is not linearly separable, it is represented as a small network rather than a single perceptron.

4 Structured-Bath Benchmark

A full thermodynamic gate under a structured reservoir should not be attempted before the non-Markovian solver is numerically stable. We therefore study a single qubit coupled to a bosonic bath with spectral density

$$J(\omega) = 2\alpha\omega^s \exp(-\omega/\omega_c). \tag{6}$$

The benchmark tracks trace preservation, Hermiticity, purity, and the final relaxation observable across sub-Ohmic, Ohmic, and super-Ohmic regimes.

Table 2: Single-qubit structured-bath benchmark.

Regime	s	Final purity	Final σ_z	Runtime (s)
Sub-Ohmic	0.5	0.5070	-0.1184	20.996
Ohmic	1.0	0.5248	0.2230	6.582
Super-Ohmic	3.0	0.8421	0.8272	4.479

The numerical-health checks are within the current tolerance: the largest trace deviation is 1.884×10^{-5} and the largest Hermiticity error is 5.011×10^{-7}. Separate time-step and memory-time scans pass a 10^{-3} final-observable tolerance. The practical working point for the next stage is

$$dt = 0.05, \qquad t_{\mathrm{mem}} = 1.0, \qquad \chi_{\mathrm{svd}} = 10^{-6}. \tag{7}$$

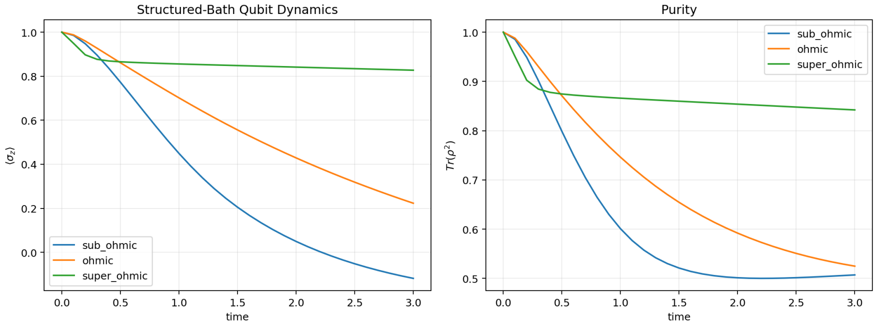

Figure 7: Reduced single-qubit dynamics for three structured-bath regimes. This benchmark provides the numerical bridge between the Markovian thermal-neuron model and later memory-bearing gate simulations.

5 Floquet-Buffered Reservoir Coupling

To reduce direct exposure of the logical qubit to the reservoir, we introduce a driven buffer between the logical subsystem and the bath,

$$S \longleftrightarrow F(t) \longleftrightarrow B, \tag{8}$$

and compare it with direct coupling $S \leftrightarrow B$. A minimal buffer Hamiltonian is

$$H_F(t) = \frac{\epsilon_F}{2}\sigma_z + A\cos(\Omega t)\sigma_x. \tag{9}$$

The principal operational metric is the trace distance between two logical output states,

$$D(t) = \frac{1}{2}\left\| \rho_S^{(0)}(t) - \rho_S^{(1)}(t) \right\|_1. \tag{10}$$

The drive cost is monitored through the work proxy

$$W_{\text{drive}} = \int_0^{t_f} \mathrm{Tr}\left[\rho(t)\frac{\partial H_F(t)}{\partial t}\right] dt. \tag{11}$$

Table 3: Direct and Floquet-buffered logical distinguishability.

Architecture	Final trace distance	Integrated drive-work proxy
Direct reservoir contact	0.1353	0.0000
Floquet-buffered contact	0.1818	0.4604

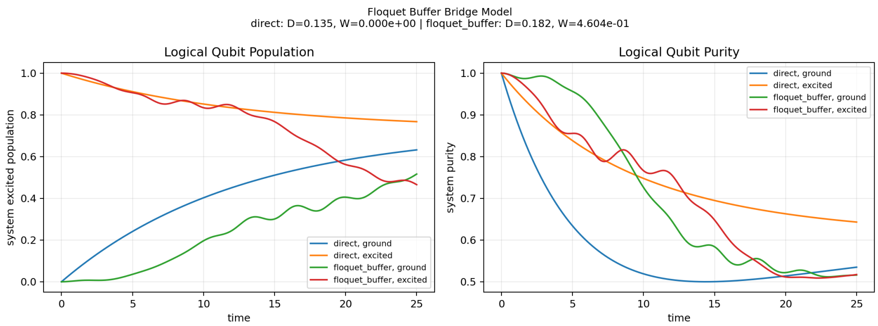

Figure 8: Logical-qubit population and purity for direct and Floquet-buffered reservoir contact. The default buffered case retains larger final distinguishability, at the cost of finite drive work.

6 Parameter Screening and Gate-Prototype Decision

A single successful Floquet-buffer run is insufficient evidence for a gate architecture. We therefore screen the drive amplitude, drive frequency, and system-buffer coupling. The screening study finds positive final trace-distance gain in 14 of 15 tested points. The largest gain occurs for a weak system-buffer coupling, with

$$\Delta D = 0.711315, \qquad W_{\text{drive}} \approx -0.019354 \tag{12}$$

in the present work-proxy convention. This value should be interpreted as a screening indicator, not as a completed thermodynamic accounting.

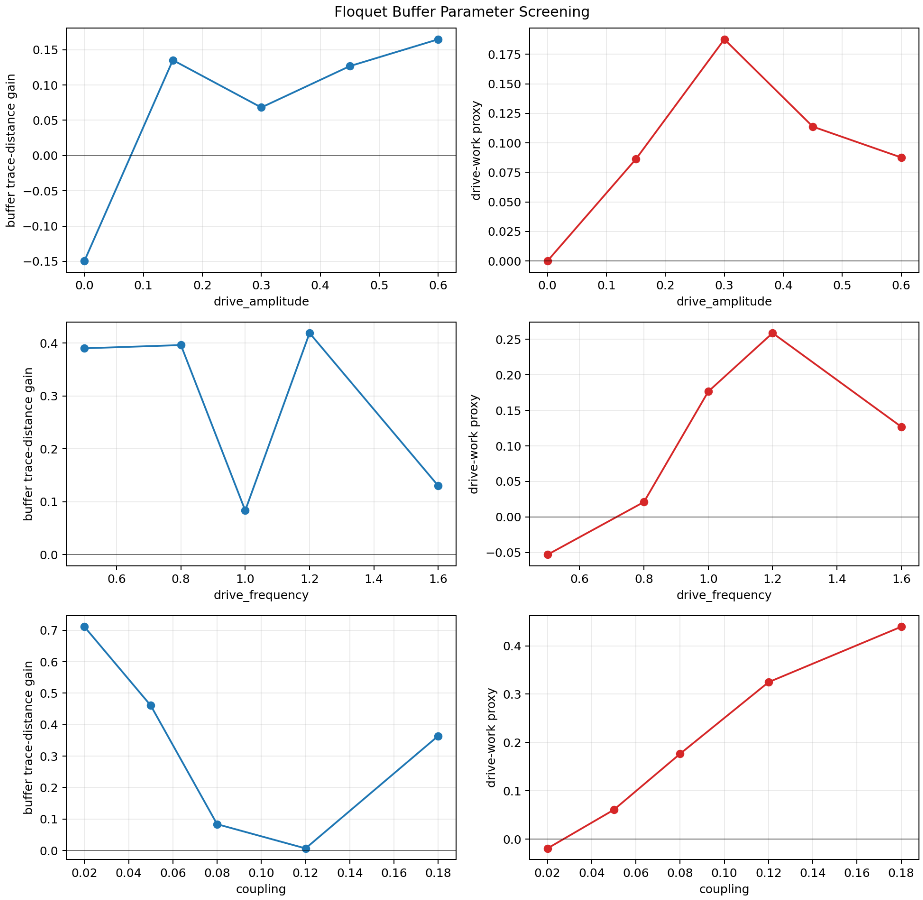

Figure 9: Floquet-buffer parameter screening. The buffer is assessed by the gain in final trace distance relative to direct reservoir contact and by the associated integrated drive-work proxy.

The decision analysis combines three requirements: numerical health of the non-Markovian benchmark, positive Floquet-buffer distinguishability signal, and a baseline NOT response sharp enough to define distinguishable logical outputs. The resulting verdict is *risky-go*: the next simulation should be a minimal NOT-style prototype, but a full three-qubit multi-bath gate remains premature.

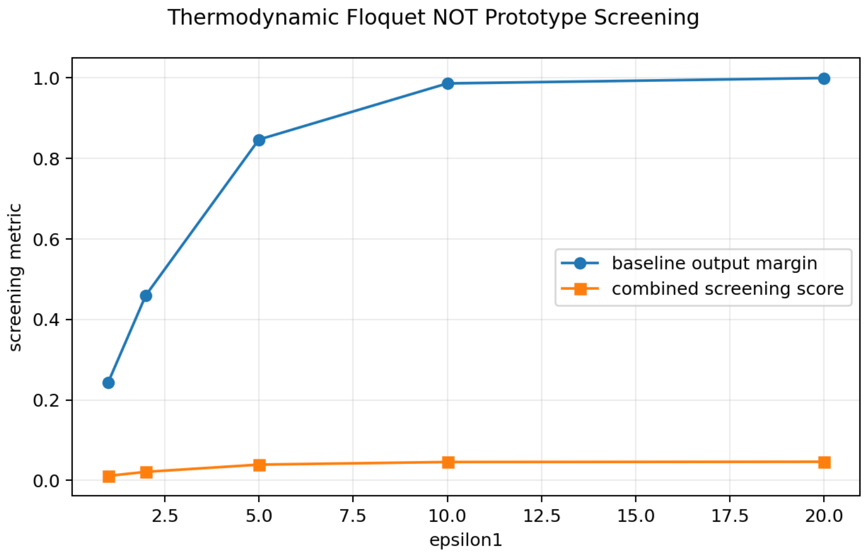

Figure 10: Screening of NOT-gate energy scale for a Floquet-buffered prototype. The largest collector energy tested gives the strongest combined score, while intermediate energy scales already provide usable margins for a first full-gate attempt.

7 Discussion

The results support the following interpretation. The Markovian reset model provides a clean thermodynamic-neuron reference: logic is obtained through virtual-temperature control and bounded reservoir relaxation. Increasing the collector energy scale improves logical sharpness but increases dissipation, reproducing the expected thermodynamic trade-off.

The structured-bath benchmark indicates that the numerical machinery is stable enough for a controlled next experiment, but it does not by itself establish a thermodynamic gate. The Floquet-buffer bridge adds an operationally meaningful signal: in the tested regimes, a driven intermediate system can increase the distinguishability of logical outputs relative to direct reservoir contact. The correct next step is therefore not a full NOR or XOR under multiple structured baths, but the smallest NOT-style experiment in which direct and buffered reservoir contact are compared under the same convergence diagnostics.

8 Conclusion

We have reconstructed the baseline thermodynamic-neuron logic model and extended the project to a Floquet-buffered reservoir architecture. The reconstructed gates establish the Markovian reference; the single-qubit structured-bath benchmark supplies numerical stability checks; and the Floquet-buffer sweep provides a first positive screening signal for improved logical distinguishability. The next scientific target is a minimal Floquet-buffered thermodynamic NOT prototype with one structured output bath. Heat, work, and entropy-production claims for the final non-Markovian gate should remain provisional until interaction energy, memory effects, and external drive work are audited consistently.

A Reproducibility Note

All figures and tables in this manuscript were generated from local Python simulations. The main reproducibility commands are:

```
scripts/run_baseline.sh
scripts/run_non_markovian_research.sh
scripts/run_dt_convergence_scan.sh
scripts/run_memory_convergence_scan.sh
scripts/run_floquet_buffer.sh
scripts/run_floquet_buffer_sweep.sh
scripts/run_project_decision.sh
scripts/run_thermodynamic_floquet_gate_prototype.sh
```

Implementation paths, generated CSV files, and run logs are intentionally kept out of the main text so that the presentation remains in the style of a research article.